

FINAL INTEGRATION READINESS

1. What is my final component?

Data Analytics — HealthConnect Appointment & No-Show Analytics Dashboard and Decision-Support Findings. The component includes the cleaned and validated appointment dataset, KPI calculations, Power BI dashboard, descriptive/diagnostic analysis, key findings, business insights and recommendations.

2. What problem does it address?

It addresses the need to understand appointment outcomes and identify patterns associated with missed appointments. The analysis helps HealthConnect understand where no-shows occur, which appointment characteristics are associated with higher observed no-show rates, and which findings may be relevant for further predictive analysis.

3. What is its current status?

Ready for final integration. The core analysis, KPI validation, dashboard and key findings have been completed. The validated results can now be incorporated into the final HC-POD presentation and used alongside the Data Science predictive component.

4. What did I improve during Week 7?

During Week 7, I validated the analytical calculations against the source data, checked dashboard outputs, refined the visualisations and reviewed the interpretation of the findings. A KPI calculation/label issue was identified and corrected: 48.46% is the validated no-show rate (2,423/5,000), while 5.26% is the cancellation rate (263/5,000). I also strengthened the interpretation of the findings by distinguishing observed associations from causal claims.

5. Which other track(s) does it depend on?

Data Science. Data Analytics depends on the Data Science track for the predictive modelling output so that descriptive findings can be compared with the predictive results and incorporated into the final integrated HealthConnect solution.

6. Which track(s) depend on my output?

Data Science depends directly on the validated analytical findings, KPI definitions and identified variables/patterns supplied by Data Analytics for model interpretation and comparison. The final HC-POD presentation also depends on these validated findings as the evidence base for the analytics section.

7. What output will I provide?

I will provide the final validated KPI results, Power BI dashboard, analytical findings, key visualisations, interpretation of important no-show patterns, business insights, recommendations and documented analytical limitations. I will also provide the relevant findings to Data Science for comparison with the predictive model.

8. What output do I need from another track?

I need the final Data Science predictive modelling output, including the relevant model findings and interpretation of important predictive features, so that the descriptive findings can be compared with the predictive results and incorporated into the final integrated solution.

9. What evidence demonstrates readiness?

Evidence includes the final Power BI dashboard, validated KPI calculations, Excel/data validation work, analytical findings, visualisations, documented Week 7 refinements, corrected KPI definitions, Data Analytics → Data Science output exchange, Data Science modelling output received, and the final HC-POD presentation materials.

10. What limitations remain?

The analysis is based on the available project dataset and therefore may not capture every factor influencing appointment attendance. The findings identify associations rather than causality. Reminder-channel differences, distance and other patterns should not automatically be interpreted as causal effects. Predictive results also represent estimated risk rather than certainty. Further real-world validation, data-quality checks, model evaluation and appropriate privacy/governance controls would be required before operational deployment.

Final Readiness Statement

The Data Analytics component is ready for final HC-POD integration. The core dataset analysis, KPI calculations, dashboard and business findings have been validated and refined. The main remaining integration dependency is the incorporation and comparison of the Data Science predictive modelling output.

The final Data Analytics contribution will provide the evidence needed to connect the historical appointment patterns with the predictive component, particularly around previous no-show behaviour, booking lead time, distance and other relevant appointment characteristics.

The component is therefore ready to move from an individual track deliverable into the integrated HealthConnect solution.